Engineering Workbench

Application Walkthrough & Screenshot Evidence

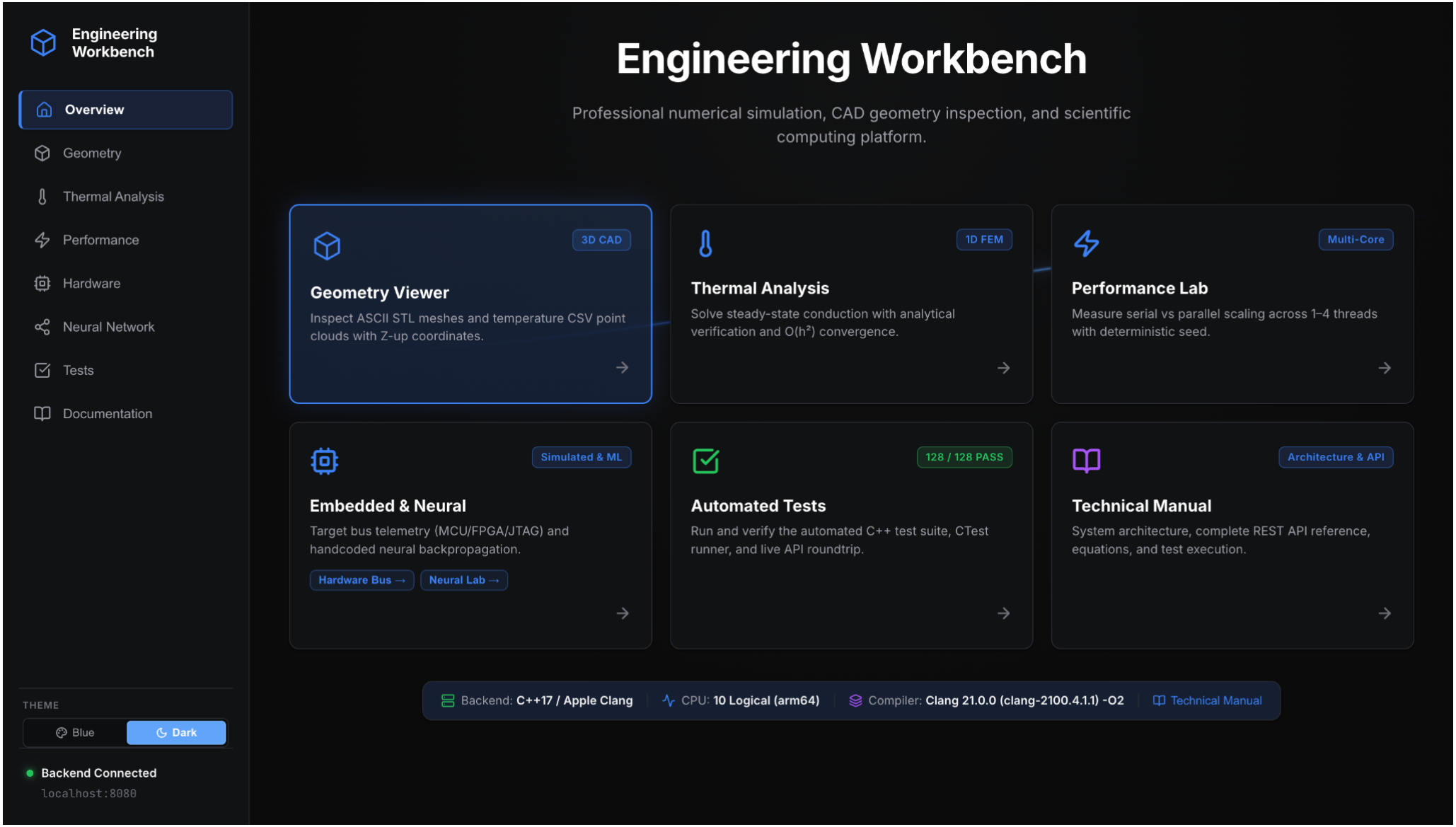
This document provides a visual walkthrough of the application's main modules. Screenshots document the interface state at capture time; they are not a substitute for raw CSV outputs, console logs, or a clean-machine Windows launch test.

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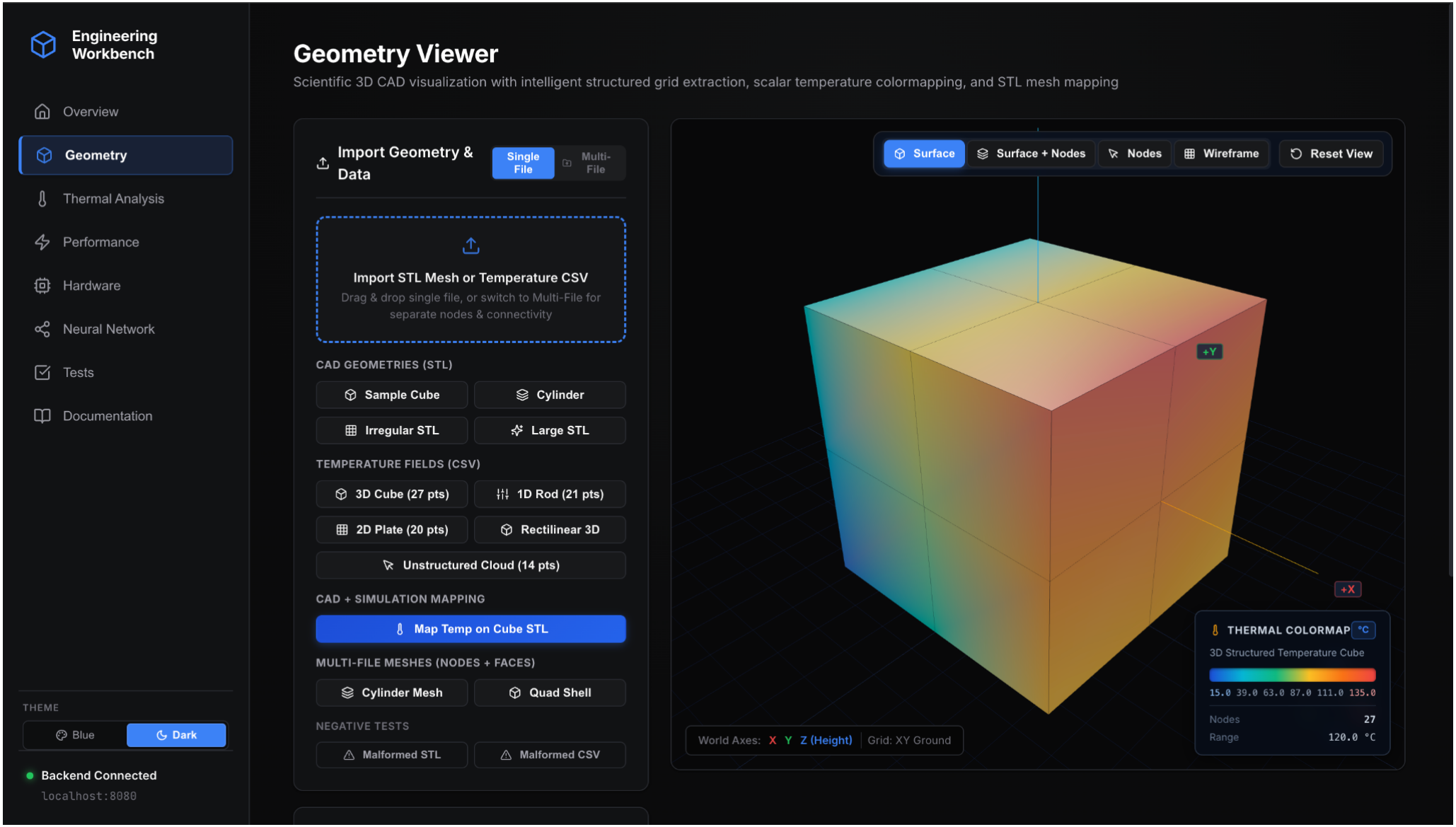
Evidence note: values in this walkthrough are transcribed from the visible UI screenshots. Where relevant, verify them against exported result files before submitting the final technical report.

Application Overview



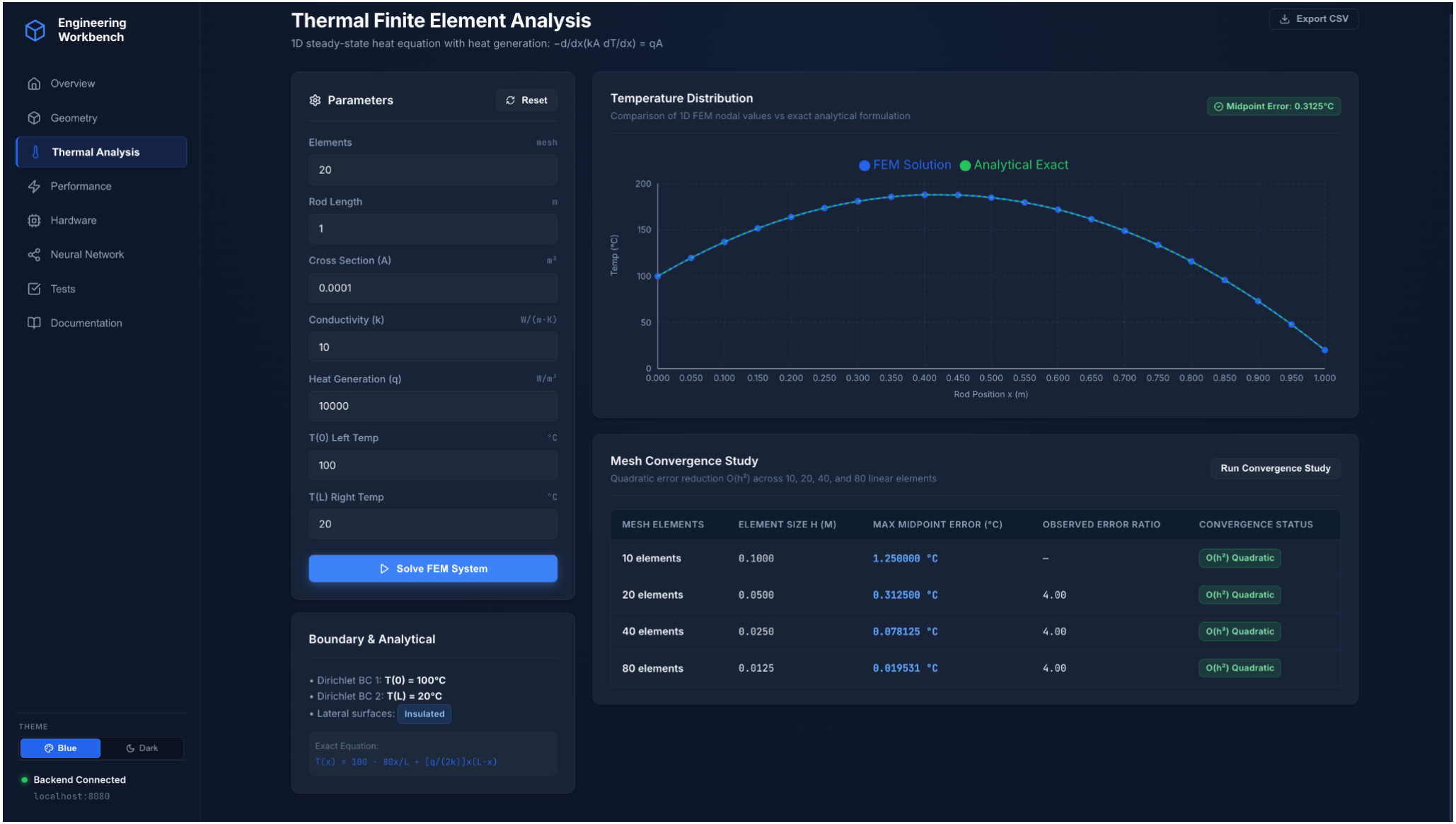
Main dashboard showing the Engineering Workbench modules and navigation.

Geometry Viewer



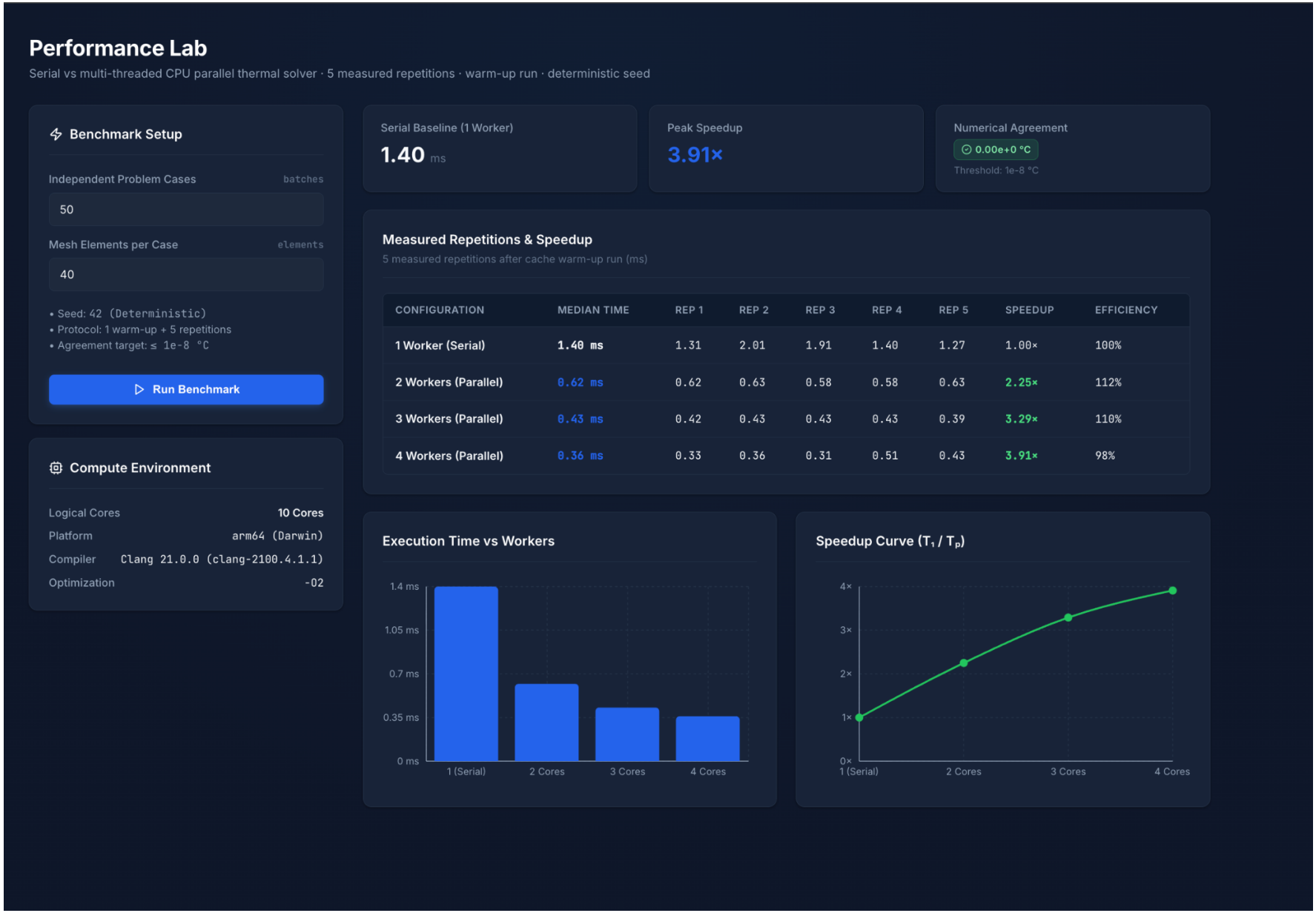
Geometry view showing a loaded 3D object and its visualization controls.

Thermal Finite Element Analysis



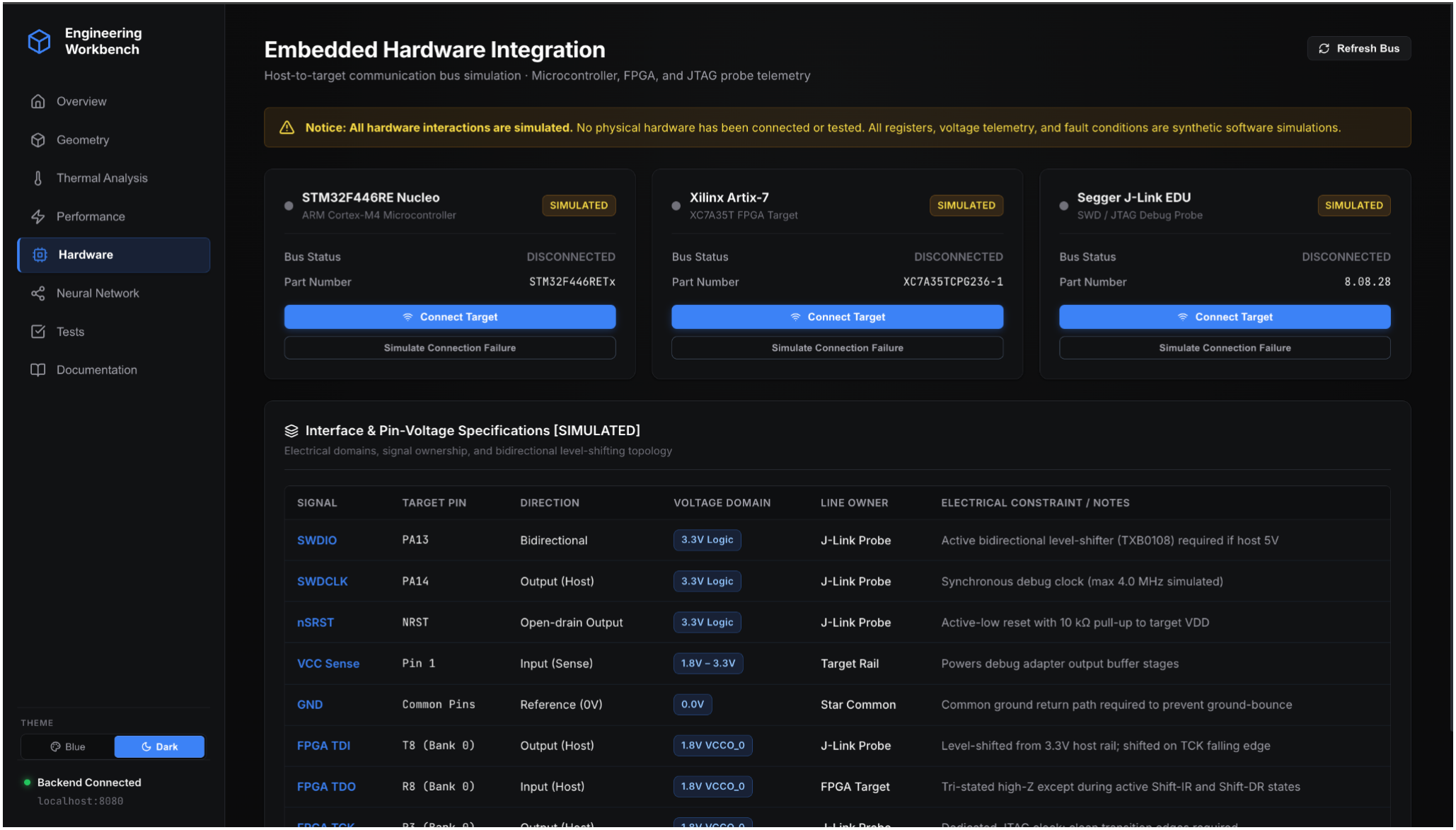
Thermal analysis screen with boundary conditions, FEM temperature distribution, and mesh-convergence results. Visible values: midpoint error 0.3125°C at 20 elements; reported errors are 1.25°C, 0.3125°C, 0.078125°C, and 0.019531°C for 10, 20, 40, and 80 elements.

Performance Lab



Serial and parallel benchmark dashboard. The UI reports a 1.40 ms serial median and 3.91× peak speedup; validate against exported raw measurements before making final claims.

Embedded Hardware Integration



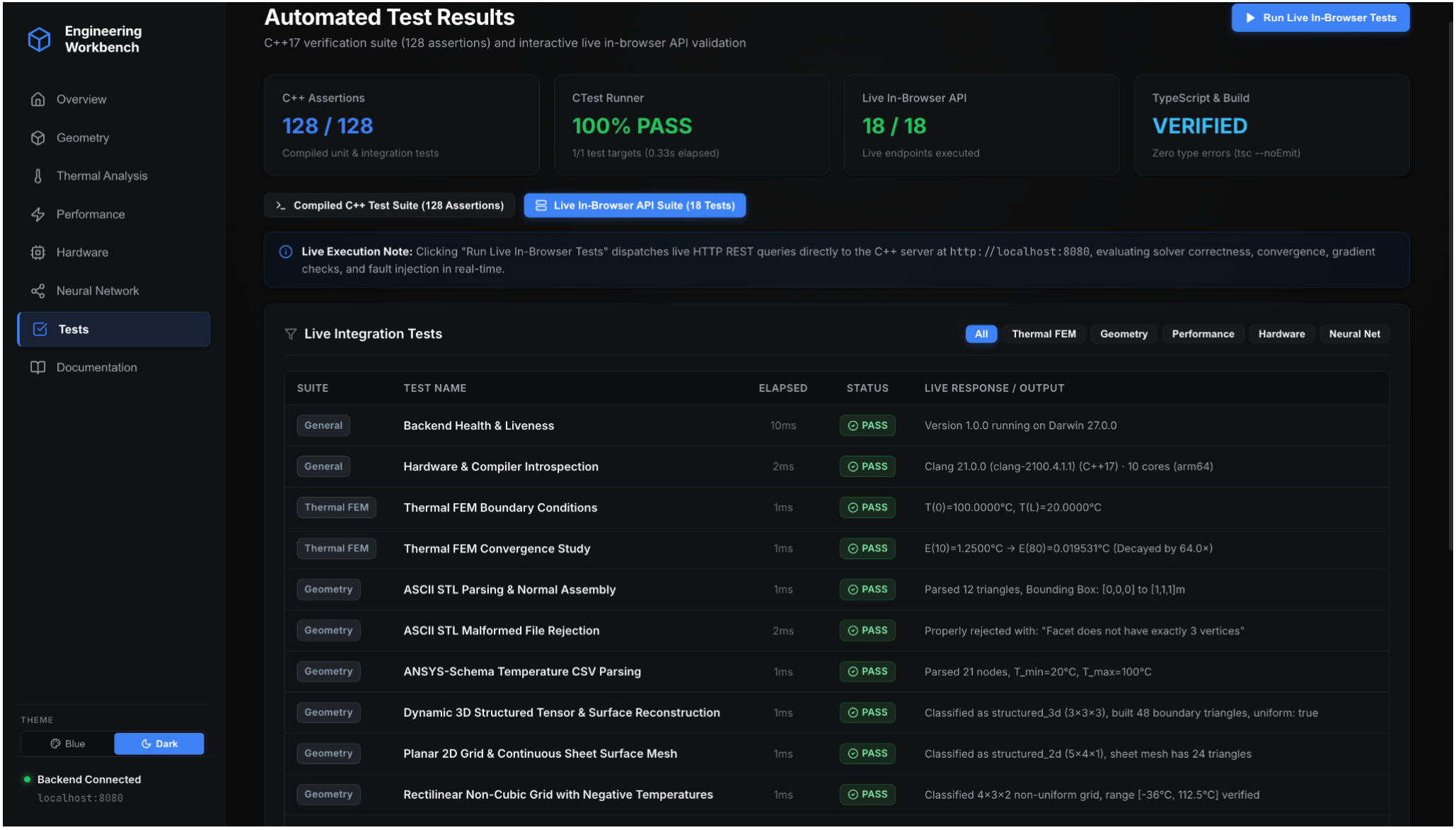
Hardware integration view showing the mock embedded adapters and their displayed status.

Neural Network Lab



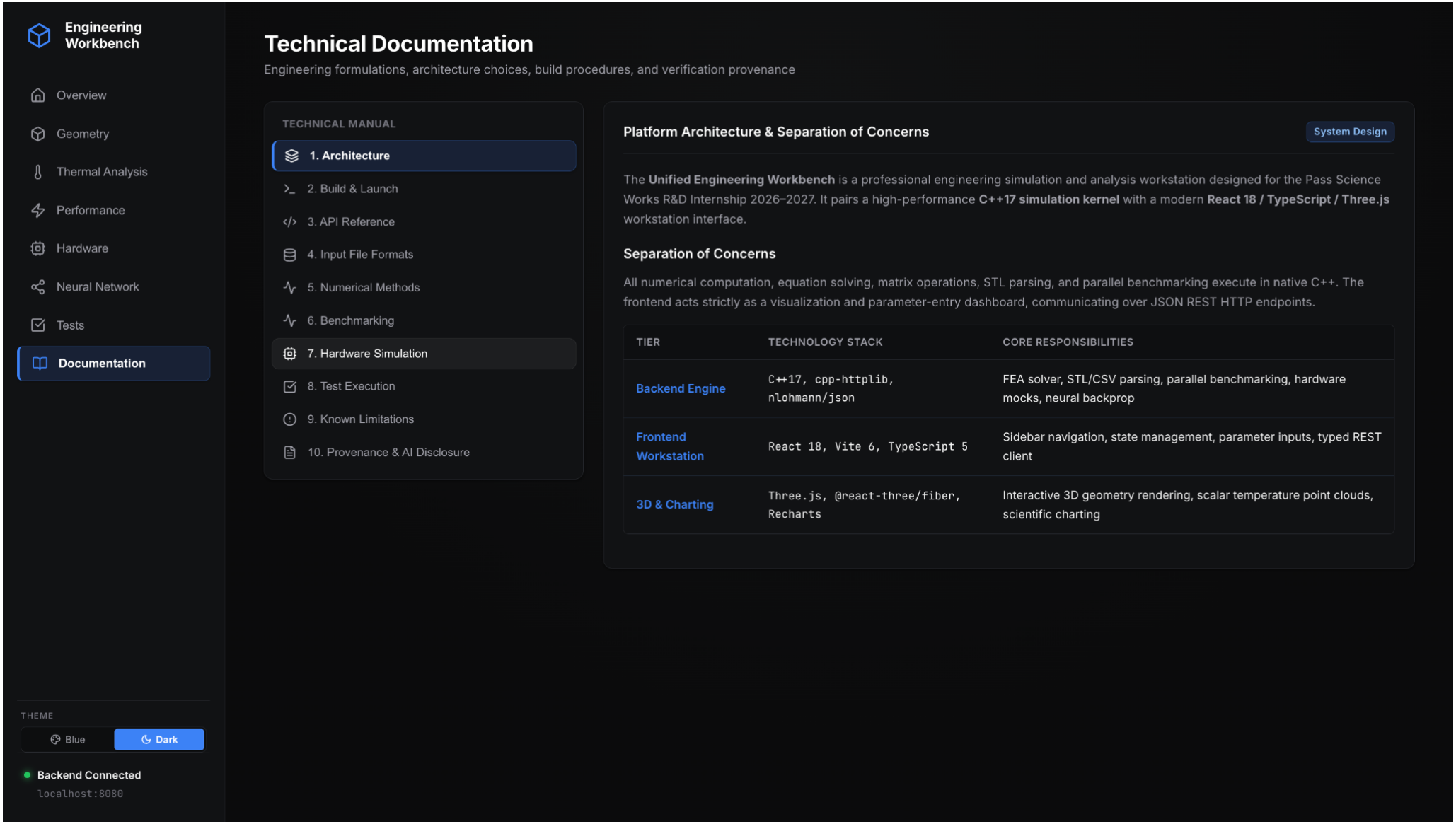
Neural-network training dashboard. The UI reports 100 iterations, learning rate 0.1, final loss 2.6830e-20, and maximum finite-difference discrepancy 1.13e-10.

Automated Test Results



Automated test results screen. The screenshot records the status displayed by the application; include the corresponding test log if available.

Technical Documentation



Documentation page within the application, showing available technical information.